The empirical recurrence for OEIS A233870: recovering a conjecture that is not written down

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Abstract

OEIS A233870 records "Empirical recurrence of order 71", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 512$ vertices and satisfies some monic recurrence of order at most S; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 71, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 71 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A233870 is "Number of (n+1)X(2+1) 0..7 arrays with every 2X2 subblock having the sum of the absolute values of all six edge and diagonal differences equal to 21.". It has offset 1 and begins

$$1176, 5404, 26488, 155104, 909408, 5878272, 36964980, 249710396, \ldots$$

The entry states:

> Empirical recurrence of order 71 (see link above)

As of the "Last modified" line on the live entry (Jun 02 2025 09:02:27, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 512$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 512$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1024$ exact terms of the model returns order 71 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{71} c_i\, a(n-i),$$

c_1	16	c_{19}	124786052970205	c_{37}	−17050303466645036062	c_{55}	−20353227062509260
c_2	70	c_{20}	174681147417858	c_{38}	33035106090355365140	c_{56}	15229030795862785
c_3	−2420	c_{21}	−1088653785919536	c_{39}	−55364020315858102223	c_{57}	51638880686177212
c_4	3931	c_{22}	−524243954791846	c_{40}	−47351400037195327701	c_{58}	−7830392353078384
c_5	151922	c_{23}	7334776406233765	c_{41}	18084025123942898663	c_{59}	−9715281878696531
c_6	−642095	c_{24}	−1147537383899276	c_{42}	54506898284519663540	c_{60}	20924352243333888
c_7	−5160051	c_{25}	−38859701382742587	c_{43}	−31275826572321963916	c_{61}	13074970656310528
c_8	34529681	c_{26}	25606984609236235	c_{44}	−50021095897105666324	c_{62}	−356007234592819
c_9	97104551	c_{27}	163080009346479065	c_{45}	38194644163685622956	c_{63}	−118805021290700
c_{10}	−1089354611	c_{28}	−180770746342631052	c_{46}	36105368553443334285	c_{64}	392978263351296
c_{11}	−632870484	c_{29}	−541227063811857175	c_{47}	−35273999268793694095	c_{65}	65515612094464
c_{12}	23133018644	c_{30}	852193349364228191	c_{48}	−20003216380298587161	c_{66}	−26922974838784
c_{13}	−17387093464	c_{31}	1401835207644359589	c_{49}	25219683816340121890	c_{67}	−1658275168256
c_{14}	−350462814496	c_{32}	−3019604561177663890	c_{50}	8094113130905332550	c_{68}	1022055284736
c_{15}	618248982966	c_{33}	−2732913141665289679	c_{51}	−14053704481190676604	c_{69}	−7596408832
c_{16}	3880951546939	c_{34}	8382039064477328627	c_{52}	−2089897079516009616	c_{70}	−16043212800
c_{17}	−10641964539879	c_{35}	3601517199710299387	c_{53}	6090952166833991504	c_{71}	855638016
c_{18}	−31308391707646	c_{36}	−18559722031986251864	c_{54}	1341884063385077728		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 71, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $71 + 4$ terms removed returns order 71 as well, so $\deg q_{\min} = 71 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $71 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 71 that this sequence satisfies — and that family turns out to have exactly one member.

References

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